SEBASTIAN L. TAGALOG

Cavite, Philippines | 0945 352 6336 | tagalogsebastian1@gmail.com
Portfolio: https://seb-lucas.github.io/
LinkedIn: www.linkedin.com/in/sebastian-tagalog-7122573b0
GitHub: https://github.com/Seb-Lucas



PROFESSIONAL SUMMARY

Computer Engineering graduate with 240 hours of manufacturing on-the-job training and hands-on experience configuring and programming a Sentry2 Camera Vision Sensor for a product inspection prototype. Experienced in hardware troubleshooting, product testing, Arduino and ESP32-based embedded systems, sensor integration, and PHP/MySQL web application development. Seeking an entry-level opportunity in technical support, IT, electronics, manufacturing, or software development.

TECHNICAL SKILLS

Hardware and Embedded Systems: Arduino Uno, Arduino Nano, ESP32, ESP8266, microcontrollers, servo motors, DC motors, sensors, HC-05 Bluetooth module, hardware interfacing
Electronics and Technical Support: Hardware troubleshooting, PC assembly, circuit integration, prototype development, equipment and hardware testing
Programming Languages: C/C++ (Intermediate), Python (Beginner), JavaScript (Beginner), PHP (Beginner)
Web Development, Databases and Tools: HTML, CSS, React (Beginner), MySQL, Firebase, Arduino IDE, MIT App Inventor, Visual Studio Code, Git, GitHub

PROFESSIONAL EXPERIENCE

On-the-Job Trainee | July 2025 – August 2025 | 240 hours

James Polymers Manufacturing Corporation | Bacoor, Cavite

- Configured and programmed a Sentry2 Vision Sensor for a product inspection prototype designed to classify company products as good or rejected based on visual inspection.
- Troubleshot hardware and tested the prototype with approximately five actual company product samples in different colors; demonstrated its classification functionality to the company owner and OJT supervisor.
- Collaborated with another OJT team on a PHP/MySQL ERP inventory module for raw material management, contributing to frontend and backend development, database tables, CRUD operations, login functionality, search, and filtering.

SELECTED PROJECTS

Arduino-Controlled Robotic Arm

Arduino Uno | Two servo motors | Joystick module

- Developed the control program for a functional robotic arm, using joystick input to control two servo motors for manual positioning and movement of up to 180 degrees.
- Programmed and integrated joystick inputs with servo motor controls, allowing coordinated arm movements and precise manual positioning.
- Tested and refined the control system to achieve functional movement and demonstrate the arm's ability to lift lightweight objects.

Bluetooth-Controlled Remote-Controlled Car

Arduino Uno R3 | HC-05 Bluetooth module | L293D motor driver IC | Two DC motors | MIT App Inventor

- Developed an Android control application using MIT App Inventor to transmit vehicle movement commands to an Arduino Uno R3 through an HC-05 Bluetooth module.
- Programmed the Arduino to interpret Bluetooth commands and control forward, reverse, left, and right movement through an L293D motor driver.
- Integrated the mobile application with the HC-05 Bluetooth module, establishing communication between the Android interface and the vehicle's control system.

Water Level Monitoring System

Arduino Uno | HC-038 water level sensor | Green, yellow, and red LEDs

- Developed a water level monitoring circuit using an HC-038 sensor and three LED indicators to display water level status.
- Programmed threshold-based sensor readings to activate green, yellow, and red LEDs, providing visual indicators for different water level conditions.
- Integrated and tested the sensor, microcontroller, and LED indicators to verify accurate status detection and corresponding visual alerts.

EDUCATION

Bachelor of Science in Computer Engineering | Graduated August 2026

Cavite State University – CCAT Campus

CERTIFICATIONS

- TESDA National Certificate II (NC II) – Computer Systems Servicing
- CompTIA A+ – AI Essentials